

BA/BSc/BCom(Vocational) Hons. in Computer Application Part-II

PAPER V

DATABASE MANAGEMANT SYSTEM-I

THEORY

Introduction to DBMS. Purpose, differences with respect to conventional file processing system. Data abstraction, data independence, database language (data-definition manipulation & control language), database users, manager & administrator. Transaction & Storage management, overall system structure .

Data model-object –base-logical model (introduction) record base logical model, Network model-Ds digrams DBTG CODASYL model TS diagram, IMS Database System, its implementation techniques. Introduction to relational model and physical data model.

Entity sets and concept, design mapping and keys .entity-relation diagram introduction to entity-relationship (E-R) model , E-R database schema tables.

Data base storage, Physical storage tertiary storage and storage access, file structure, file organization, records in files data-dictionary storage and storage structure security and integrity.

PRACTICAL

Foxpro release 2.6 for windows

Introduction to foxpro, interface and FOXPRO commands, creating database, saving adding, editing data records and database file. Browsing data, power tool RQBE, dialogue screen design, report and form, design, PRG files under FOXPRO.

Books recommended:

1. DATABASE SYSTEM CONCEPT-KORTH AND SILBERSCHATZ
2. INTRODUCTION TODATABASE MANAGEMENT-NAVEEN PRAKASH
3. MASTRING FOXPRO 2.6-CHARLES SEIJEL
4. FOXPRO 2.5 FOR DOS AND WINDOWS-TAXALI
5. INTRODUCTION TO DATABASE MANAGEMANT SYSTEM-MAJMDAR
6. DATABASE MANAGEMANT SYSTEM-POST
7. DBASE III PLUS MADE SIMPLE WITH FOXPRO- TAXALI

PAPER VI

STRUCTURAL PROGRAMMING

THEORY

Concept of structural programming, structure of a programming, primitive & non-primitive data, union, pointers, dynamic memory location, function, data structure, linear and non-linear data structures, arrays, algebra of matrix, sparse matrix stack queue circular queue, linked lists, tree, graph, path, sorting and searching techniques. Algorithms, rate of growth, time analysis, big-notation, related to insertion, deletion, insertion at a point, deletion from a point etc.

PRACTICAL

C Programming

Structure of a program, variables, constants, modifiers, escape sequences, expressions, and operators. Formatted and standard input - output. Control flow (if-else, if-else if, do while, while, switch-case, for use of break and continue.) String/character handling, one and two dimensional arrays, functions and arguments, passing parameters, recursive functions, structures, sorting and searching. Pointers, linked lists, files; preprocessor, error handling.

COBOL Programming:

Structure of a COBOL program, PIC clauses, verbs and arithmetic verbs. I/O statements (Accept, display, move), control and looping constructs (if, coto, perform, condition-name-condition.) Rename redefine, group data identifiers, filter cause. Arrays, sort, search and index verb. File handling in Cobol, sequential, random and ISAM files.

BOOKS RECOMMENDED :

1. DATA STRUCTURE - LIPSCHUTZ
2. DATA STRUCTURE- TREEMBLEY AND SORENSON
3. PROGRAMMING IN C - GOTTFRIED
4. PROGRAMMING IN C- BALAGURUSWAMY
5. COBOL - ROY, GHOSH, DASTIDAR
6. STRUCTURED COBOL- KAZMIER AND PHILIPPAKIS

PAPER VII

COMPUTER GRAPHICS

THEORY

Graphics, overview of graphics devices, elementary graphics, points, line circle, ellipse and curves, functions and algorithms, pixel addressing, mathematics for computer graphics, co-ordinate, points, vectors and matrices. Why graphics. Uses and advantages. Various types of graphics programs. Drafting packages. DTP packages. Hardware requirements, text generation fonts, page layout and its programs, graphics for DTP, creating bitmap, vector graphics. Forms generation, design principles, and documents design. Presentation and printing.

PRACTICAL

Ms – power point:

Power point interface, menus commands and toolbars, creating presentation using wizard, creating new presentation, creating presentation from template, working with charts, working in outline view, organization charts and tables, working on presentations, working in slide sorter view, making slide shows.

Page maker:

Page maker interface, Menus commands and toolbars, Creating a publication, placing and formatting of text, Spacing, Indent, Tabs, Chapter and page numbering, Editing publication, Working with graphs and layers, Working with names and color, Text wrap and reversing.

Corel draw:

Corel draw interface, menus commands and toolbars, working with shapes and objects, draw rectangle, square, circle, ellipse, stars, spiral, rotate and fill color to any shape, working with outlines and file, working with curves and bitmaps, edit , trace. Mode resample and inflate of bitmaps, special effects, blending and envelope tools.

BOOKS RECOMMENDED:

1. INTRODUCTION TO COMPUTER GRAPHICS- KRISHNAMURTHY
2. COREL DRAW – RAMAESH BANGINA
3. POWER POINT- RAMESH BANGINA
4. PAGE MAKER – RAMESH BANGINA
5. DTP – M.C.SHARMA
6. PRINCIPAL OF INTERACTIVE COMPUTER GRAPHICS -NEWMAN

PAPER VIII

OBJECT ORIENTED PROGRAMMING

THEORY

Concept of OOPS and differences with procedural languages , characteristics of OPPS , idea of objects, class, data abstraction and encapsulation, inheritance polymorphism, dynamic binding I/O stream , can, coot, I/O manipulation.

Data types, operators, control structure and looping statements, functions arrays. **Objects and classes:** classes and objects, constructor, destructor, overloaded constructor, overloading. **Operator overloading :** overloading unary operator, overloading binary operators, and data conversion.

Inheritance: derived classes and base class, protected access specified, derived class constructors, class hierarchies, abstract base class, public and private inheritance, multiple inheritance, containership.

Pointer : Address and pointers , pointers and arrays, memory management, NEW & DELETE pointer to objects, linked list, pointer to pointer, THIS pointer.

Virtual function : Virtual function, fiend function, static function .

Files and streams: string I/O, object I/O, I/O with multiple objects file pointer. Error handling, redirection.

PRACTICAL

Object oriented programming using C++.

BOOKS RECOMMENDED:

1. OOPS CONCEPT- BOOCH
2. PROGRAMMING WITH C++ - KANETKER
3. MASTERING IN C++ - LAFORE
4. C++ - BALAGURUSWAMY
5. PROGRAMMING WITH C++ - ANDEREDS